

Metaphyseal Chondrodysplasia, Jansen Type (JMC): Comprehensive Disease Characterization

A mechanistic and clinical knowledge-base report. Evidence sources: human clinical, mouse model, in vitro, and computational/structural. Ontology term suggestions (HPO, GO, CL, UBERON, CHEBI, NCIT) are provided where applicable.

Summary

Metaphyseal Chondrodysplasia, Jansen type (JMC; OMIM #156400) is an ultra-rare autosomal dominant skeletal dysplasia caused by **heterozygous gain-of-function (activating) missense mutations in *PTH1R*** (the parathyroid hormone / PTH-related peptide type 1 receptor; HGNC:*PTH1R*, gene OMIM 168468, UniProt Q03431). *Five recurrent mutations at three residues are established causes — H223R (by far the most common), T410P, T410R, I458R, and I458K — each of which renders the receptor constitutively active in a ligand-independent manner, producing basal Gsα–cAMP signaling in the absence of hormone. This single molecular lesion explains the entire disease: it simultaneously delays growth-plate chondrocyte hypertrophy (yielding metaphyseal widening, bowed limbs, and severe short stature) and drives PTH-independent hypercalcemia** — a biochemical signature nearly unique among skeletal dysplasias and central to diagnosis.

The pathophysiology is now well understood as the mirror image of loss-of-function *PTH1R* disease. Constitutively active *PTH1R* inhibits cellular **salt-inducible kinases (SIK2/SIK3)**, de-repressing **class IIa histone deacetylases** and phenocopying the skeletal program of combined *Sik2/Sik3* deletion. Within the normal **PTHrP–Indian hedgehog (Ihh) feedback loop**, *PTH1R* signaling — mediated critically by **Gsα** — restrains chondrocyte hypertrophy; when this signal is "locked on," maturation is pathologically delayed. Downstream renal effects include hypercalciuria, hypophosphatemia (with reported FGF23 elevation), nephrocalcinosis, and, in some patients, progressive chronic kidney disease. JMC is therefore best framed as a **dose-dependent constitutively-active *PTH1R* disorder** whose severity tracks with the intrinsic constitutive activity of the specific mutation (H223R severe; T410R milder).

There is no approved disease-specific therapy. Management is symptomatic — hydration, dietary calcium restriction, bisphosphonate plus thiazide for hypercalciuria, and orthopedic surgery for limb deformities. The leading mechanism-directed strategy is a class of **PTH1R inverse-agonist peptides** that suppress the mutant receptor's basal signaling and partially rescue skeletal defects in mouse models; a synthetic PTH inverse agonist is in clinical development. A key knowledge gap: essentially all molecular-profiling data derive from **mouse and cell models**, not human omics datasets, reflecting the disorder's ultra-rarity and tissue inaccessibility.

1. Disease Information

Overview. JMC is a rare form of short-limbed dwarfism (metaphyseal chondrodysplasia) characterized by severe growth-plate abnormalities with metaphyseal widening/irregularity, progressive skeletal deformity, distinctive craniofacial features, and — uniquely among the metaphyseal chondrodysplasias — **PTH-independent hypercalcemia** with low/normal circulating PTH and PTHrP.

Key identifiers:

Resource	Identifier
OMIM (disease)	#156400
Gene	<i>PTH1R</i> (aka PTHR1, PTH/PTHrP receptor); OMIM 168468; HGNC:PTH1R; UniProt Q03431*
MONDO	Metaphyseal chondrodysplasia, Jansen type (Mendelian; MONDO ID not resolved in this investigation)
Category	Mendelian, autosomal dominant

Synonyms / alternative names: Jansen metaphyseal chondrodysplasia; Jansen-type metaphyseal dysplasia; Jansen's disease; metaphyseal dysostosis, Jansen type; Murk Jansen type metaphyseal chondrodysplasia.

Source of information. Because JMC is ultra-rare, the knowledge base derives from **aggregated disease-level resources** (OMIM, Orphanet) and **published case reports / small case series** rather than large EHR cohorts. The largest natural-history dataset assembled to date comprised 24 patients ([PMID: 29788189](#)).

2. Etiology

Primary cause — genetic. JMC is caused by **heterozygous activating (gain-of-function) missense mutations in *PTH1R***. In HEK293/COS-7 reporter cells the mutant receptors exhibit constitutive, ligand-independent cAMP accumulation. Disease is **autosomal dominant**; most cases arise **de novo**, though multigenerational transmission (e.g., a father and two sons with T410R) is documented (Finding F001; [PMID: 29788189](#); [PMID: 27160269](#)).

"Five different activating PTH/PTH-related peptide (PTHrP) receptor (PTHR1) mutations have been reported as causes of Jansen metaphyseal chondrodysplasia (JMC), a rare disorder characterized by severe growth plate abnormalities and PTH-independent hypercalcemia." — [PMID: 29788189](#)

Genetic risk factors. The causal variants themselves are the risk factors; there are no known common susceptibility loci or modifier alleles in humans. Because the mechanism is a dominant, activating point mutation, the mutation is both necessary and sufficient.

Environmental risk factors. None established. There are no known toxic, infectious, occupational, or lifestyle causes. Family history is relevant only in the minority of inherited (non-de novo) cases.

Protective factors. No genetic or environmental protective factors have been identified.

Gene–environment interactions. None established for disease causation. Environmental modulation is limited to **symptom management** (e.g., dietary calcium restriction, hydration reducing the hypercalcemia/hypercalciuria burden).

3. Phenotypes

JMC is a **multisystem disorder** with a defining skeletal + mineral-ion phenotype plus renal, ocular, and possibly cardiovascular involvement (Findings F002, F009, F012).

Skeletal / physical manifestations (clinical signs; congenital-to-childhood onset; progressive)

- **Metaphyseal widening and irregularity** of long bones; disorganized growth-plate cartilage. *HPO: HP:0003025 Metaphyseal widening; HP:0000944 Abnormality of the metaphysis.*
- **Short-limbed short stature / dwarfism** — adult heights well below the 3rd percentile in all patients **except** those with the milder T410R mutation (F010; [PMID: 29788189](#)). *HPO: HP:0004322 Short stature.*
- **Bowed limbs / progressive limb deformity**; most patients undergo orthopedic surgery. *HPO: HP:0002751 Kyphoscoliosis; HP:0002986 Radial bowing.*
- **Craniofacial dysmorphism** (prominent supraorbital ridges, midface hypoplasia, micrognathia). *HPO: HP:0000316 Hypertelorism; HP:0000347 Micrognathia.*

Mineral-ion / laboratory abnormalities (biochemical; childhood-predominant)

- **PTH-independent hypercalcemia** (normal at birth; elevated ~0.15–10 yr, mean 11.8 ± 1.37 mg/dL; tends to normalize in adults, 10.0 ± 1.03 mg/dL). *HPO: HP:0003072 Hypercalcemia. CHEBI:29108 calcium(2+).*
- **Hypercalciuria** (urinary Ca/creatinine children 0.80 ± 0.40 ; adults 0.28 ± 0.19). *HPO: HP:0002150 Hypercalciuria.*
- **Hypophosphatemia** (lower end of age-specific ranges); reported **FGF23 elevation**. *HPO: HP:0002148 Hypophosphatemia. CHEBI:43474 hydrogenphosphate.*
- **Low / normal PTH** despite hypercalcemia — the diagnostic hallmark.

"Postnatal calcium levels were normal in most patients, but elevated between 0.15 and 10 years (11.8 ± 1.37 mg/dL) and tended to normalize in adults (10.0 ± 1.03 mg/dL)." — [PMID: 29788189](#)

Renal (progressive)

- **Nephrocalcinosis** (most patients), nephrolithiasis, and **advanced chronic kidney disease** in a subset (2/24). HPO: HP:0000121 Nephrocalcinosis; HP:0012622 Chronic kidney disease.

"Most patients with JMC had undergone orthopedic surgical procedures, most had nephrocalcinosis, and two had advanced chronic kidney disease." — [PMID: 29788189](#)

Other systems

- **Ocular findings** characterized in JMC (calcium-deposition context; [PMID: 39108358](#)).
- **Severe infantile hypertension** proposed as a previously unrecognized feature ([PMID: 31977144](#)). HPO: HP:0000822 Hypertension.

"Hypertension has not been previously associated with JMC." — [PMID: 31977144](#)

- **Hypercalcemia symptom complex in infants:** failure to thrive, poor feeding, constipation, polyuria, irritability, lethargy, seizures, hypotonia ([PMID: 33990852](#)).

"Hypercalcaemia presents clinically with a range of symptoms including failure to thrive, poor feeding, constipation, polyuria, irritability, lethargy, seizures and hypotonia." — [PMID: 33990852](#)

Quality-of-life impact. Substantial: severe short stature and progressive limb deformity requiring repeated orthopedic surgery impair mobility and function; nephrocalcinosis/CKD add chronic renal morbidity. No disease-specific EQ-5D/SF-36 data are available (ultra-rare disease).

Onset / severity / progression / frequency (summary):

Phenotype	HPO	Onset	Severity	Progression	Frequency
Short stature	HP:0004322	Childhood	Severe (except T410R)	Progressive	Nearly all
Metaphyseal widening	HP:0003025	Congenital/ childhood	Severe	Progressive	All
Hypercalcemia	HP:0003072	Infancy– childhood	Variable	Peaks in childhood, normalizes in adults	Most
Hypercalciuria	HP:0002150	Childhood	Moderate– severe	Persistent	Consistent
Nephrocalcinosis	HP:0000121	Childhood	Variable	Progressive	Most
CKD	HP:0012622	Later	Severe	Progressive	Subset (2/24)

4. Genetic / Molecular Information

Causal gene: *PTH1R* (chromosome 3p21.31), encoding a class B1 (secretin-family) G-protein-coupled receptor.

Pathogenic variant spectrum (all heterozygous, activating):

Variant	Location	Frequency in largest cohort	Constitutive activity	Phenotype
H223R	1st intracellular loop (exon M2)	18/24 patients (most common)	High	Severe
T410P	TM helix	Single case	Moderate–high	Severe
T410R	TM helix	Father + 2 sons	Lower	Milder
I458R	TM7	Single case	~8× basal vs WT	Severe-like
I458K	TM7	Single case	High	Severe

(F001; [PMID: 29788189](#); [PMID: 10487664](#))

"The H223R mutation occurred in 18 patients. T410P, I458R and I458K each occurred in single cases; T410R was present in a father and his two sons." — [PMID: 29788189](#)

- **Variant classification (ACMG/AMP):** pathogenic (functional constitutive-activation evidence + segregation). Some newly reported *PTH1R* variants are classified VUS (e.g., p.E465K, [PMID: 37840415](#)).
- **Variant type:** all missense.
- **Allele frequency:** absent from population databases (gnomAD) — private/de novo pathogenic variants.
- **Origin: germline** (de novo in most; inherited in some pedigrees). Germline mosaicism is plausible but not systematically documented.
- **Functional consequence: gain of function** — constitutive, ligand-independent Gsα–cAMP signaling (F001; [PMID: 27160269](#)).

"Constitutive, ligand-independent cAMP accumulation was observed in HEK293T cells expressing the Mut-PTH1R." — [PMID: 27160269](#)

Allelic disorder spectrum (differential diagnosis; F007). *PTH1R* variants cause a spectrum defined by signaling **direction** and **zygosity**:

Disorder	Zygoty	Effect	Skeletal consequence
JMC	Heterozygous	Gain of function (constitutive)	Delayed ossification + hypercalcemia
Blomstrand lethal chondrodysplasia	Homozygous / compound-het	Severe loss of function	Neonatal-lethal, accelerated ossification
Eiken syndrome	Homozygous	Milder LOF	Delayed ossification
PHP-like (hypocalcemia/hyperphosphatemia)	Homozygous (e.g., R186H)	LOF	Mineral-ion disturbance
Primary failure of tooth eruption (PFE)	Heterozygous	LOF	Non-syndromic dental
Brachydactyly type E syndrome (H8 variants E465K, E469K)	Heterozygous	Impaired signaling	Mild short stature, dental anomalies

"Heterozygous PTH1R mutations that lead to constitutively activity were identified in Jansen metaphyseal chondrodysplasia, and homozygous or compound heterozygous mutations that lead to less-active or completely inactive receptors were identified in patients with Blomstrand lethal chondrodysplasia." — [PMID: 10912527](#)

"Severe loss-of-function homozygous mutations in PTH1R are incompatible with life as in Blomstrand's lethal chondrodysplasia, characterized by accelerated growth plate ossification." — [PMID: 40904804](#)

Modifier genes / epigenetics / chromosomal abnormalities. No human modifier genes are established. Downstream **class IIa HDACs** are key epigenetic effectors of PTH1R signaling in bone (mechanistic, mouse; see §6). No chromosomal abnormalities are involved (JMC is a point-mutation disorder).

5. Environmental Information

Not applicable as a cause. JMC is a purely monogenic, constitutively-active-receptor disorder. No environmental factors, lifestyle factors, or infectious agents cause or trigger the disease. Environmental variables are relevant only to **symptom modulation** (dietary calcium, fluid intake affecting stone/nephrocalcinosis risk).

6. Mechanism / Pathophysiology

Causal chain (upstream → downstream)

Heterozygous activating PTH1R mutation (H223R / T410P/R / I458R/K)
 | (stabilizes active receptor conformation)

▼
 Constitutive, ligand-independent Gsα activation

▼
 Elevated basal cAMP → PKA → inhibition of SIK2/SIK3

▼
 de-repression of class IIa HDACs / CRTC

▼
 GROWTH PLATE:
 delayed chondrocyte
 hypertrophy; prolonged
 hypertrophic zone;
 disorganized metaphysis

▼
 Transcriptional program of
 constitutively-active PTH1R
 (= phenocopy of Sik2/Sik3 loss)

▼
 Metaphyseal widening, bowed limbs, short stature

KIDNEY / MINERAL: constitutive renal PTH1R signaling
 → ↑ Ca reabsorption / bone Ca release → HYPERCALCEMIA (PTH-independent)
 → hypercalciuria, ↑FGF23 → hypophosphatemia
 → nephrocalcinosis → CKD

Molecular pathways

- **PTH/PTHrP receptor → Gsα → adenylyl cyclase → cAMP → PKA** is the primary driver. PTH1R also couples to **Gq/PLC**, but the JMC phenotype is dominated by the Gsα–cAMP arm ([PMID: 15765186](#); [PMID: 40571720](#)).
- **cAMP–SIK–HDAC axis**: PTH1R activation inhibits SIKs; combined *Sik2/Sik3* deletion phenocopies constitutively active PTH1R, with **class IIa HDACs** as the key regulated substrates in chondrocytes and osteocytes (F003; [PMID: 31430259](#)).

"Combined deletion of *Sik2* and *Sik3* in osteoblasts and osteocytes led to a dramatic increase in bone mass that closely resembled the skeletal and molecular phenotypes observed when these bone cells express a constitutively active PTH1R that causes Jansen's metaphyseal chondrodysplasia." — [PMID: 31430259](#)

"genetic evidence demonstrated that class IIa histone deacetylases were key PTH1R-regulated SIK substrates in both chondrocytes and osteocytes" — [PMID: 31430259](#)

Developmental mechanism — the PTHrP–Ihh loop (F006)

In normal endochondral development, **PTHrP** (made in the periarticular growth plate) acts on PTH1R to **keep chondrocytes proliferating and delay hypertrophy**; **Ihh** (from prehypertrophic chondrocytes) induces PTHrP, forming a negative-feedback loop that sets the pace of differentiation and the site of bone-collar formation. **Gsα is the critical mediator**: chondrocyte-specific *Gsα* knockout **accelerates** hypertrophy (phenocopying receptor knockout), whereas constitutive PTH1R activation (JMC) does the opposite — **delaying** hypertrophy.

"As chondrocytes go through a program of proliferation and then further differentiation into post-mitotic, hypertrophic chondrocytes, PTHrP action keeps chondrocytes proliferating and delays their further differentiation." — [PMID: 16831900](#)

"These results show that G(s)alpha negatively regulates chondrocyte differentiation and is the critical signaling mediator of the PTH/PTH-rP receptor in growth plate chondrocytes." — [PMID: 15765186](#)

Downstream effectors delaying hypertrophy/apoptosis include **Bcl-2 upregulation**, suppression of **p57** and **Runx2**, **SOX9 phosphorylation**, and upregulation of **Zfp521** (a rescue target — see §15).

"PTHrP increases the expression of Bcl-2, a protein that controls programmed cell death in several cell types, in growth plate chondrocytes both in vitro and in vivo, leading to delays in their maturation towards hypertrophy and apoptotic cell death." — [PMID: 9008714](#)

Cellular processes / tissue damage

- **Delayed chondrocyte hypertrophic differentiation** and prolonged persistence of hypertrophic chondrocytes (growth-plate disorganization).
- **High-turnover bone disease**: histomorphometry in H223R children shows irregular architecture, increased osteoid, prolonged osteoid maturation, intense cortical osteoclast activity, marrow fibrosis, and osteocytes with osteoid buildup in lacunae/shortened canaliculi — a **PTH-like bone phenotype** (F009; [PMID: 39950977](#)).

"Cortical bone displayed areas of intense osteoclast activity and scattered marrow fibrosis." — [PMID: 39950977](#)

Protein dysfunction / structural basis (F011)

PTH1R is a **prototypical class B1 GPCR** coupling to both Gs and Gq (UniProt Q03431). Cryo-EM structures with PTH, PTHrP, abaloparatide, LA-PTH, and M-PTH(1-14) show the agonist N-terminus engaging the **transmembrane bundle** to drive Gas activation ([PMID: 37148874](#); [PMID: 40571720](#)). The JMC mutations map to **activation-critical regions**: H223 in the first intracellular loop near the G-protein interface, and T410/I458 in transmembrane helices — positions where substitutions **stabilize the active conformation**, producing ligand-independent signaling.

"we describe cryo-EM structures of the PTH1R in complex with fragments of the two hormones, PTH and PTH-related protein, the drug abaloparatide, as well as the engineered tool compounds, long-acting PTH (LA-PTH) and the truncated peptide, M-PTH(1-14)" — [PMID: 37148874](#)

"The critical N terminus of each agonist engages the transmembrane bundle in a topologically similar fashion, reflecting similarities in measures of Gas activation." — PMID: 37148874

Metabolic / immune / molecular profiling

- **Metabolic:** dysregulated calcium/phosphate homeostasis (hypercalcemia, hypophosphatemia, hypercalciuria; FGF23 elevation contributing to renal phosphate wasting).

"Hypophosphatemia is also a hallmark of JMC, and recently, increased fibroblast growth factor 23 (FGF23) levels have been reported in this syndrome." — PMID: 22278430

- **Immune involvement:** none established.
- **Molecular profiling (F013):** No large-scale human transcriptomic/proteomic/metabolomic/single-cell datasets specific to JMC exist in public repositories (GEO/ArrayExpress/PRIDE/MetaboLights). The molecular signature comes from **mouse/cell models** — the SIK–HDAC transcriptional program (PMID: 31430259) and the Col2a1-H223R growth-plate delayed-maturation program (PMID: 9391087). Human-level "omics" is limited to targeted biochemistry and bone histomorphometry/IHC.

"immunohistochemical analysis demonstrated increased in PTH1R expression in both osteoblasts and fibroblastic cells on the bone surface" — PMID: 39950977

Suggested GO terms: GO:0007189 (adenylate cyclase-activating GPCR signaling), GO:0002062 (chondrocyte differentiation), GO:0003416 (endochondral bone growth), GO:0071107 (response to parathyroid hormone), GO:0007050 (cell cycle arrest). **Suggested CL terms:** CL:0000138 (chondrocyte), CL:0000743 (hypertrophic chondrocyte), CL:0000062 (osteoblast), CL:0000137 (osteocyte), CL:0000092 (osteoclast).

7. Anatomical Structures Affected

Organ / system level (primary → secondary): - **Skeletal system (primary):** long-bone metaphyses, growth plates (physes), cranium, mandible, spine. *UBERON:0002515 metaphysis; UBERON:0002481 growth plate cartilage / physis; UBERON:0001474 bone element.* - **Urinary system (secondary):** kidney (nephrocalcinosis, CKD). *UBERON:0002113 kidney.* - **Ocular (secondary):** cornea/eye (calcium deposition). *UBERON:0000970 eye.* - **Cardiovascular (candidate):** systemic vasculature (infantile hypertension). *UBERON:0001981 blood vessel.* - **Endocrine / mineral-homeostasis axis** (functionally central).

Tissue / cell level: - **Cartilage / growth-plate chondrocytes** (proliferative and hypertrophic zones). *CL:0000138 chondrocyte; CL:0000743 hypertrophic chondrocyte.* - **Bone: osteoblasts, osteocytes, osteoclasts** (high-turnover remodeling). *CL:0000062 osteoblast; CL:0000137 osteocyte; CL:0000092 osteoclast.*

Subcellular level: plasma membrane (receptor); cytoplasmic cAMP/PKA signaling; nucleus (HDAC/CRTC-regulated transcription). *GO:0005886 plasma membrane; GO:0005737 cytoplasm; GO:0005634 nucleus.*

Localization / lateralization: skeletal involvement is **generalized and bilateral/symmetric** (systemic germline receptor activation).

8. Temporal Development

Onset: congenital-to-early-childhood; radiographic metaphyseal changes present early; serum calcium **normal at birth**, then rises between ~0.15 and 10 years (F002). Onset pattern is **chronic/insidious**.

Progression: - Skeletal deformity is **progressive** through childhood; short stature is established and permanent (except milder T410R). - **Hypercalcemia** peaks in childhood and **tends to normalize in adulthood** — a distinctive age-dependent course ([PMID: 29788189](#)). - **Renal disease** (nephrocalcinosis → CKD) is progressive in a subset.

Disease course: chronic, lifelong. No spontaneous remission of the skeletal phenotype; the hypercalcemia component partially self-attenuates with age.

Critical periods / windows of opportunity: the **growth-plate-active childhood window** is the key period for skeletal morbidity and the theoretical window for mechanism-directed (inverse-agonist) intervention before physal closure.

9. Inheritance and Population

Epidemiology. JMC is **ultra-rare** — only a few dozen well-documented patients worldwide (largest cohort n=24; [PMID: 29788189](#)). Precise prevalence/incidence are not established (Orphanet: <1/1,000,000).

Inheritance: autosomal dominant. Most cases are **de novo**; vertical transmission occurs (e.g., T410R father→sons; H223R mother→sons).

Penetrance / expressivity: high penetrance but **variable expressivity** — an H223R-carrier mother "was never overtly hypercalcemic and was therefore not diagnosed until her sons were found to be affected" (F010; [PMID: 27410178](#)). Incomplete penetrance has been reported for some novel *PTH1R* variants ([PMID: 37840415](#)).

"the now 38-year-old mother was never overtly hypercalcemic and was therefore not diagnosed until her sons were found to be affected by JMC" — [PMID: 27410178](#)

Genotype-phenotype correlation (F010): severity tracks with the **degree of constitutive activity**. H223R → profound hypercalcemia + severe short stature; **T410R** → **milder** (only mutation with near-normal adult height; near-normal longevity in humanized mice).

"Adult heights were well below the 3rd percentile for all patients, except for those with the T410R mutation." — [PMID: 29788189](#)

Anticipation / mosaicism / founder effects / consanguinity / carrier frequency: No genetic anticipation (not a repeat-expansion disorder). No founder effects; consanguinity is **not** relevant (dominant, de novo). Carrier frequency is not applicable. (Consanguinity is relevant instead to the *recessive* allelic disorders — Blomstrand, Eiken.)

Demographics: no ethnic predilection; no strong sex bias reported (dominant, sporadic). Presents in infancy/childhood.

10. Diagnostics

Biochemical (cornerstone): the diagnostic signature is **hypercalcemia + hypercalciuria + hypophosphatemia with LOW/NORMAL PTH** (PTH-independent hypercalcemia). FGF23 may be elevated. This distinguishes JMC from PTH-dependent hypercalcemias (F002, F012). *LOINC: serum calcium, phosphate, intact PTH; urine calcium/creatinine ratio.*

Imaging: skeletal radiographs show **metaphyseal widening/irregularity, cupping, sclerotic and lucent changes**, and progressive deformity — the defining radiographic features (case reports [PMID: 10901979](#); [PMID: 6974367](#)). Renal ultrasound demonstrates nephrocalcinosis.

Genetic testing (confirmatory): targeted **single-gene sequencing of *PTH1R*** or skeletal-dysplasia/hypercalcemia **gene panels**; WES/WGS in undiagnosed cases. Detection of a heterozygous activating variant (H223R, T410P/R, I458R/K) is confirmatory. Chromosomal microarray/karyotype are not informative (point-mutation disorder).

Bone biopsy / histomorphometry: high-turnover changes (increased osteoid, osteoclast activity, marrow fibrosis; increased PTH1R IHC) — research/selected use ([PMID: 39950977](#)).

Differential diagnosis (F007, F012): - Other **metaphyseal chondrodysplasias** — Schmid, McKusick, Spahr, Shwachman — separable by clinical/radiographic/biochemical criteria; **only JMC has hypercalcemia** ([PMID: 8644413](#)). - **Rickets** (a classic radiographic mimic). - **PTH-dependent hypercalcemias:** neonatal severe hyperparathyroidism, familial hypocalciuric hypercalcemia (FHH) — these have **high PTH** ([PMID: 33990852](#)). - **Other PTH-independent pediatric hypercalcemias:** Williams-Beuren syndrome, hypophosphatasia, idiopathic infantile hypercalcemia ([PMID: 34774247](#)).

"Other types are Schmid, Spahr, McKusick, Schwachman and Jansen, which can be separated by clinical, radiographic, genetic and biochemical criteria." — [PMID: 8644413](#)

"Hypercalcemia can rarely be associated with immobilization, genetic diseases in children such as Williams-Beuren syndrome, Hypophosphatasia, Jansen Metaphyseal Chondrodysplasia (JMC)" — [PMID: 34774247](#)

Screening: No population newborn screening. **Cascade genetic testing** of at-risk relatives is appropriate in familial cases; prenatal/preimplantation testing is feasible once the family variant is known.

11. Outcome / Prognosis

Survival / mortality: JMC is generally **compatible with survival into adulthood** (contrast with neonatal-lethal Blomstrand chondrodysplasia). No formal life-expectancy tables exist; prognosis is dominated by **renal** (CKD from nephrocalcinosis) and **orthopedic** morbidity.

Morbidity / function: substantial lifelong disability from **severe short stature and progressive limb deformity** (most patients require orthopedic surgery), plus chronic renal disease in a subset. Hypercalcemia-related symptoms burden infancy/childhood.

Disease course / complications: nephrocalcinosis, nephrolithiasis, progressive CKD (2/24 advanced), skeletal deformity requiring surgery, and candidate infantile hypertension.

Prognostic factors: the **specific mutation** is the strongest predictor — **T410R milder**, H223R more severe. The natural attenuation of hypercalcemia in adulthood is a favorable biochemical trend.

12. Treatment

No approved disease-specific therapy exists. Management is **symptomatic/supportive** (F004, F008).

Supportive medical management: - Hydration; **dietary calcium restriction**; control of hypercalcemia/hypercalciuria. - **Bisphosphonate (alendronate)** reduced hypercalciuria in an adult H223R patient; **normocalciuria required adding a thiazide diuretic**; FGF23 remained normal under treatment ([PMID: 22278430](#)). *NCIT: bisphosphonate therapy; thiazide diuretic.*

"Treatment with alendronate reduced hypercalciuria; however, normocalciuria was only obtained with the association of thiazide diuretic." — [PMID: 22278430](#)

Orthopedic surgery: frequently required for limb deformity/correction. *NCIT: orthopedic surgical procedure.*

Rehabilitation / renal care: physical therapy; nephrology management of nephrocalcinosis/CKD.

Mechanism-directed / experimental — PTH1R inverse agonists (leading strategy; F004): - **N-terminally truncated PTH/PTHrP antagonist peptides act as inverse agonists**, reducing the high basal cAMP of JMC mutant receptors in vitro; [L11,dW12,W23,Y36]PTHrP(7-36) reduced basal cAMP for all five mutants ([PMID: 31693237](#)). - An **inverse-agonist ligand partially rescued skeletal defects** in a JMC mouse model ([PMID: 31693237](#)). - A **backbone-modified long-acting peptide** functions as an inverse agonist of PTH1R-H223R **in vitro and in vivo** ([PMID: 38417010](#)). - A **synthetic PTH inverse agonist (PTH-IA)** is in clinical development, with anti-drug-antibody assays already established ([PMID: 42572867](#)).

"we found that certain N-terminally truncated PTH and PTHrP antagonist peptides function as inverse agonists and thus can reduce the high rates of basal cAMP signaling exhibited by the mutant PTHR1s of JMC in vitro" — [PMID: 31693237](#)

"a peptidic PTH1R inhibitor that displays prolonged activity as an antagonist of wild-type PTH1R and an inverse agonist of the constitutively active PTH1R-H223R mutant both in vitro and in vivo" — [PMID: 38417010](#)

Personalized medicine: because severity and receptor pharmacology are mutation-specific, an inverse agonist confirmed active against **all five mutants** supports a genotype-agnostic-yet-mechanism-targeted approach. Structure-guided design also extends to **de novo GPCR-targeting miniproteins** ([PMID: 42168559](#)).

13. Prevention

- **Primary prevention:** not possible for de novo dominant mutations. In **familial** cases, **genetic counseling**, prenatal diagnosis, and preimplantation genetic testing can prevent recurrence.
- **Secondary prevention:** early biochemical detection (Ca/Pi/PTH) and imaging in symptomatic infants; early nephrology surveillance to limit nephrocalcinosis/CKD.
- **Tertiary prevention:** manage hypercalciuria (bisphosphonate + thiazide) to reduce stone/nephrocalcinosis burden; orthopedic monitoring; renal protection.
- **Counseling:** genetic counseling is central — clarifying **AD inheritance, high de novo rate, and variable expressivity** (a mildly affected/undiagnosed parent may carry the variant; [PMID: 27410178](#)).

- **Immunization / public health / environmental interventions:** not applicable.
-

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *PTH1R* is highly conserved. Functional models exist in **mouse** (*Mus musculus*, NCBI Taxon 10090); the ortholog is *Pth1r*. Zebrafish and other vertebrates possess conserved PTH1R signaling.
 - **Natural disease in animals:** No well-established naturally-occurring JMC-equivalent (constitutively-active PTH1R) disease is documented in companion animals or wildlife in this investigation (OMIA not confirmed here).
 - **Comparative biology:** the PTHrP–Ihh–PTH1R growth-plate circuit is **evolutionarily conserved**; mouse models faithfully reproduce the human causal chain (see §15), making JMC a strong example of conserved endochondral-development mechanisms.
 - **Transmission:** not applicable (non-infectious, genetic).
-

15. Model Organisms

JMC is unusually well-modeled in the mouse, providing the mechanistic backbone of the field (F005).

Model	Type	Key phenotype	Reference
Col2a1-HKrk-H223R transgenic	Constitutively active receptor under $\alpha 1(\text{II})$ collagen promoter (chondrocyte-targeted)	Delayed mineralization; decelerated proliferative→hypertrophic conversion; prolonged hypertrophic chondrocytes	PMID: 9391087
Col1a1-H223R transgenic	Osteoblast-targeted	Bone phenotype (↑ cortical bone) mirroring JMC	PMID: 11713230
Humanized H223R-hPTH1R knock-in	Knock-in	Early lethal; cannot breed; substantially delayed growth-plate chondrocyte maturation	PMID: 37808400 ; PMID: 40455993
Humanized T410R-hPTH1R knock-in	Knock-in (milder allele)	Near-normal longevity/breeding; misshapen long bones, expanded metaphyses, disarrayed growth-plate zones, reduced primary spongiosa	PMID: 40455993
Sik2/Sik3 double KO	Conditional KO (phenocopy)	Bone phenotype closely resembling constitutively active PTH1R	PMID: 31430259
Zfp521 deletion in Jansen chondrocytes	Genetic rescue	Restores chondrocyte differentiation; partial rescue of bone length	PMID: 21642473
Chondrocyte-specific Gsa KO	Conditional KO (opposite phenotype)	Accelerated hypertrophy (mirror of JMC)	PMID: 15765186

"The targeted expression of constitutively active PTH/PTHrP receptors led to delayed mineralization, decelerated conversion of proliferative chondrocytes into hypertrophic cells in skeletal segments that are formed by the endochondral process, and prolonged presence of hypertrophic chondr[ocytes]." — [PMID: 9391087](#)

"The long bones of T410R mice are markedly misshapen and have expanded metaphyses with disarrayed chondrocyte zones in growth plates and reduced primary spongiosa." — [PMID: 40455993](#)

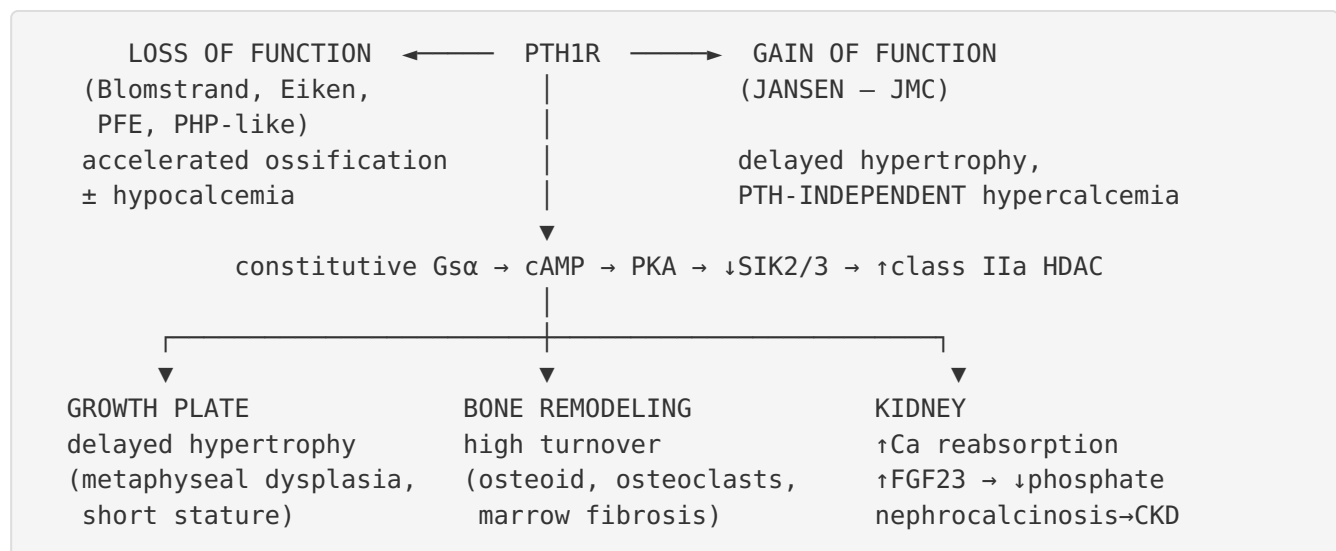
"Its ablation from Jansen chondrocytes restored normal cell differentiation, thus initiating chondrocyte apoptosis at the chondro-osseous junction, leading to partial rescue of endochondral bone formation shown by proper bone length." — [PMID: 21642473](#)

Phenotype recapitulation: Excellent for the growth-plate/skeletal phenotype and for genetic-rescue proof-of-concept (Zfp521; inverse agonists). **Limitations:** the severe H223R humanized model is early-lethal and cannot breed, complicating long-term/therapeutic studies; the milder T410R model was engineered specifically to enable these. Renal/mineral-ion aspects and human natural history (adulthood normalization of calcium) are only partially captured.

Model applications: dissecting the PTHrP-Ihh-Gs α -SIK-HDAC pathway; testing **inverse-agonist therapeutics** in vivo; studying genotype-severity relationships. **Resources:** MGI (mouse); model lines maintained by originating academic laboratories.

Mechanistic Model / Interpretation

JMC is best understood as a "**receptor stuck ON**" **disorder** — the biochemical and developmental inverse of Blomstrand chondrodysplasia (receptor OFF). A single class of lesion (heterozygous activating *PTH1R* missense mutation) produces the entire multisystem phenotype through **constitutive Gs α -cAMP signaling**:



Upstream vs downstream: the mutation (upstream) → constitutive cAMP (proximal effector) → SIK inhibition / HDAC de-repression (transcriptional node) → cell-type-specific outputs (downstream): delayed chondrocyte hypertrophy in cartilage, high-turnover remodeling in bone, and altered mineral handling in kidney. **Cell types:** growth-plate chondrocytes (proliferative + hypertrophic), osteoblasts, osteocytes, osteoclasts, and renal tubular cells. Therapeutic logic follows directly: **inverse agonists** that suppress the receptor's basal activity restore signaling toward normal and partially rescue the skeletal phenotype in mice — the most promising path to a disease-modifying treatment.

Evidence Base

PMID	Title (abbrev.)	Supports
7701349	Constitutively active mutant PTH-PTHrP receptor in JMC	Landmark 1995: H223R → constitutive cAMP (F008)
29788189	Progression of mineral ion abnormalities (n=24)	Mutation spectrum; mineral-ion natural history (F001, F002, F009, F010)
27160269	Characterization of a PTH1R missense mutation	Constitutive ligand-independent cAMP (F001)
10487664	Novel PTH1R mutation (I458R)	I458R ~8× basal cAMP; H223R most frequent (F001)
31430259	SIKs dictate PTH1R action in bone	SIK–HDAC downstream mechanism; molecular signature source (F003, F013)
16831900	PTHrP and skeletal development	Normal PTH1R role in delaying hypertrophy (F006)
15765186	Chondrocyte Gsa knockout	Gsa is critical PTH1R mediator (F006)
9008714	Bcl-2 downstream of PTHrP	Bcl-2 delays chondrocyte maturation (F006)
9391087	Targeted constitutively active receptors	Original Col2a1-H223R mouse model (F005, F013)
21642473	Deletion of Zfp521	Genetic rescue of Jansen growth plate (F005)
40455993	Humanized JMC mouse model	H223R vs milder T410R knock-ins (F005, F010)
37808400	Delayed maturation in humanized mice	Growth-plate delay in humanized model (F003, F005)
31693237	Inverse agonist rescues skeletal defects	Inverse agonists suppress mutant signaling (F004)
38417010	Backbone-modified long-acting inverse agonist	H223R inverse agonist in vivo (F004)
42572867	ECL bridging assay for PTH inverse agonist	Clinical development of PTH-IA (F004)
22278430	Alendronate + thiazide in adult JMC	Symptomatic hypercalciuria management; FGF23 (F002, F008)

PMID	Title (abbrev.)	Supports
39950977	Bone abnormalities beyond chondrodysplasia	High-turnover histomorphometry; PTH1R IHC (F009, F013)
31977144	Severe hypertension in JMC	Candidate cardiovascular feature (F009)
39108358	Ocular findings in JMC	Ocular involvement (F009)
10912527	Role of PTHrP and Ihh	GOF vs LOF dichotomy (F007)
40904804	Human diseases from homozygous PTH1R mutations	Blomstrand LOF contrast (F007)
8644413	Metaphyseal chondrodysplasia vs rickets	Differential diagnosis (F007)
33990852	Genetic neonatal/infantile hypercalcaemia	JMC in hypercalcemia differential (F012)
34774247	Rare causes of hypercalcemia	JMC as rare hypercalcemia cause (F012)
27410178	H223R with/without overt hypercalcemia	Variable expressivity (F010)
37148874	Peptide agonist engagement with PTH1R	Cryo-EM structural basis (F011)
40571720	G-protein coupling preference cryo-EM	Class B1 GPCR structure (F011)
42168559	De novo design of GPCR-targeting miniproteins	Structure-guided therapeutic design (F011)

Evidence source types: human clinical (case series/reports, biochemistry, histomorphometry), model organism (mouse transgenics/knock-ins/rescues), in vitro (HEK293/COS-7 cAMP reporter assays), and computational/structural (cryo-EM, de novo design).

Limitations and Knowledge Gaps

1. **No human omics datasets (F013).** The molecular signature of JMC is inferred from mouse (SIK–HDAC) and cell models, not human transcriptomic/proteomic/metabolomic/single-cell data — a direct consequence of ultra-rarity and inaccessible growth-plate tissue.
 2. **Small n / no formal epidemiology.** Prevalence, incidence, precise sex ratio, and life-expectancy tables are not established; the largest cohort is 24 patients.
 3. **Newer/candidate features under-characterized.** Ocular findings and infantile hypertension are reported in single/few patients and need replication before being considered core features.
 4. **Therapeutics are pre-approval.** Inverse-agonist efficacy is demonstrated in vitro and in mouse models with only partial skeletal rescue; human efficacy, dosing window (before physeal closure), and immunogenicity remain to be established.
 5. **MONDO ID not resolved** in this investigation; OMIM #156400 is the authoritative identifier used.
 6. **Renal/cardiovascular mechanisms** (FGF23 elevation, hypertension) are incompletely dissected relative to the skeletal phenotype.
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Proposed Follow-up Experiments / Actions

1. **Clinical trial of a PTH1R inverse agonist** in JMC patients (building on [PMID: 42572867](#), [PMID: 38417010](#)), prioritizing dosing during the growth-active childhood window, with skeletal, mineral-ion, and renal endpoints.
2. **Patient-derived iPSC → chondrocyte/organoid models** to generate the first **human** JMC transcriptomic/proteomic signatures and validate the SIK–HDAC axis in human cells.
3. **Longitudinal natural-history registry** capturing genotype (H223R vs T410R vs others), skeletal growth, renal function trajectory, and the adulthood normalization of hypercalcemia — powering genotype–phenotype and prognostic modeling.
4. **Systematic characterization of the renal and cardiovascular phenotypes** (FGF23 biology, nephrocalcinosis progression, infantile hypertension) in the milder T410R humanized mouse, which permits long-term/breeding studies ([PMID: 40455993](#)).

5. **Structure-guided next-generation inverse agonists / de novo miniproteins** ([PMID: 42168559](#)) optimized against all five mutant receptors, using the available cryo-EM structures ([PMID: 37148874](#)).
6. **Resolve MONDO mapping** and reconcile Orphanet/OMIM/ICD identifiers for knowledge-base ingestion.

Report compiled from 13 confirmed findings and 43 reviewed papers across 5 investigation iterations. Evidence sources span human clinical case series, mouse genetic models, in vitro receptor pharmacology, and cryo-EM structural biology.

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